Q-band ART: Adaptive Band Suppression for Anti-Rollover Tracking of Semi-Trailer Tank Trucks

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Abstract—Semi-trailer tankers are prone to rollover because lateral acceleration excites both suspension roll and liquid sloshing. Existing anti-rollover tracking methods usually introduce liquid states, roll states, or lateral load transfer ratio predictions into model predictive control, which increases dependence on liquid calibration, trailer-side sensing, and online model dimension. This paper proposes Q-band ART (Anti-Rollover Tracking), an adaptive band-suppression framework that penalizes the predicted lateral-acceleration energy in rollover-critical frequency bands. A reduced cart-roll-sloshing model explains why the critical band should be identified from the coupled acceleration-to-risk response, and the resulting Q-band term is embedded into a nonlinear model predictive tracking controller customized for semi-trailer tank trucks with acceleration smoothing, soft band constraints, preview-adaptive weights, extended-state disturbance compensation, hitch-angle-based parameter scheduling. The method is validated in simulations and 1:14 model-truck experiments, reducing peak rollover-risk proxy by xxx% while maintaining acceptable tracking error and real-time performance. Open-source implementation is available at https://github.com/KomasaQi/HitchOpen-THICV-Stack

Index Terms—Anti-rollover control, semi-trailer tanker, liquid sloshing, path tracking, model predictive control, frequency-band suppression.

I. Introduction

SEMI-TRAILER tank trucks are indispensible for liquid hazardous-chemical logistics, but their partially filled tanks make rollover a severe safety risk. Accident reports show that tank truck rollovers are often triggered by sharp steering, emergency avoidance, curved-road overspeed, or driver error, and may lead to deadly secondary disasters [1]–[3]. Because safety standards leave practical operation across partial filling rates [4], [5], free-surface sloshing must be considered in autonomous tank truck tracking.

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Studies show that with free-surface, driving excitations generate additional sloshing, making medium filling rates yielding stronger dynamic instability than either empty or fully filled cases. And the sloshing in tank has been modeled by quasi-static centroid-shift methods, mechanical equivalent models (MEMs), computational fluid dynamics, and experiments. [6]–[10]. Among the methods, MEMs especially pendulum models are widely used for their ability to balance interpretability and real-time computation, although the dominant sloshing modes still require accurate calibration [11]–[14].

For semi-trailer tank trucks, rollover prevention is further complicated by tractor-trailer articulation. High-fidelity models may include lateral, yaw, and roll motion of tractor and trailer,respectively, as well as liquid states [15]–[18]. Such models are useful offline, but online use is limited by trailer replacement, non-unified trailer interfaces, limited trailer-side sensing, and uncertain liquid/-suspension parameters. Active suspension, steering, and differential braking have all been studied for rollover prevention [19]–[21]; tank truck-specific controllers have used PID, LQR, MPC, model-free adaptive control (MFAC), and other robust control with MEMs [22]–[27].

Most existing controllers still rely on explicit liquid states, roll states, or lateral load transfer ratio (LTR) predictions. This is not always suitable for autonomous semi-trailer tank trucks with limited trailer sensing and embedded computing resources. Model predictive control (MPC) can combine tracking, actuator, acceleration, and safety constraints [28], but high-dimensional states and dense constraints challenge real-time implementation [29]. This paper therefore suppresses rollover excitation from the input side: the predicted lateral-acceleration sequence is projected onto vehicle-liquid critical bands, and the resulting energy is penalized and softly constrained. The workflow is shown in Fig. 1.

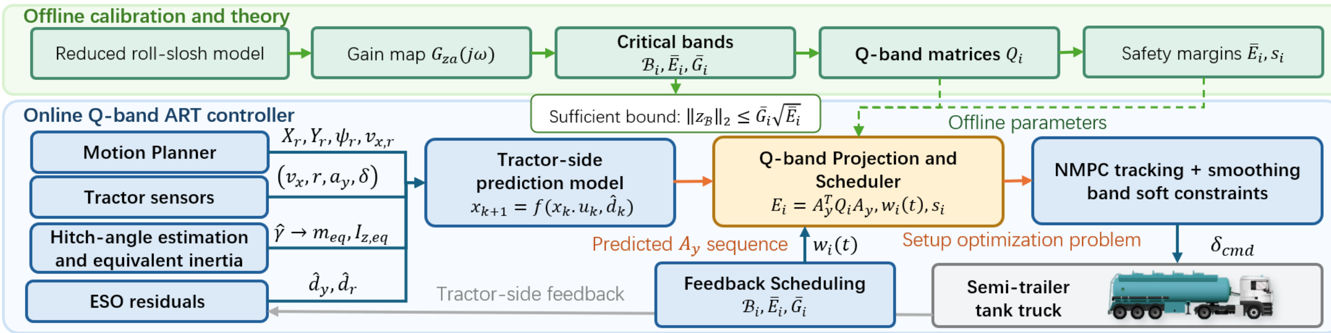

Fig. 1: Architecture of Q-band ART. Offline calibration identifies the critical bands, energy thresholds, Q-band matrices, and conservative margins. The online controller then combines tractor-side prediction, ESO residual compensation, hitch-angle-based equivalent-parameter scheduling, Q-band projection, and NMPC to output steering commands using tractor-side sensing and precomputed Q-band parameters.

The contributions are threefold. First, a Q-band anti-excitation view is introduced for semi-trailer tank truck tracking, avoiding mandatory online liquid-state modeling and observation. Second, an multi-point NMPC tracking controller with disturbance compensation and trailer-free modeling is designed specifically for semi-trailer. Third, the Q-band term is integrated predictively into the controller for anti-rollover purpose and validated through simulation and 1:14 model-truck experiments. The rest of the paper presents the model, controller, validation, and conclusions.

II. Q-Band Suppression Theory

A. Reduced Roll-Sloshing Excitation Model

Online full liquid-state prediction requires liquid-state observation [14], which is not always available. To avoid full state prediction, a reduced model is established to explain why input-side band suppression is sufficient for anti-rollover control and how dangerous frequency components are identified. The unsprung lateral motion is represented by a lateral cart, the sprung mass by a roll pendulum, and the dominant liquid mode by a pendulum attached to the tank body. Taking lateral acceleration a_y as input, the coupled roll-sloshing dynamics can be linearized as

$$\mathbf{M}\ddot{\mathbf{x}}_r + \mathbf{C}\dot{\mathbf{x}}_r + \mathbf{K}\mathbf{x}_r = \mathbf{B}_a a_y, \quad \mathbf{x}_r = [\phi, \theta]^{\mathrm{T}}, \tag{1}$$

where ϕ is the tank-body roll angle, θ is the relative liquid sloshing angle, $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ are the equivalent inertia, damping, and stiffness matrices, and $\mathbf{B}_a$ maps the base lateral acceleration into the roll-slosh coordinates. The off-diagonal terms in $\mathbf{M}$ and $\mathbf{K}$ describe the coupled free mode between suspension roll and liquid motion as

$$\det(\mathbf{K} - \omega^2\mathbf{M}) = 0. \tag{2}$$

Therefore, the dangerous frequency is not the isolated liquid natural frequency or the isolated roll frequency; the two modes interact and generate a broad peak.

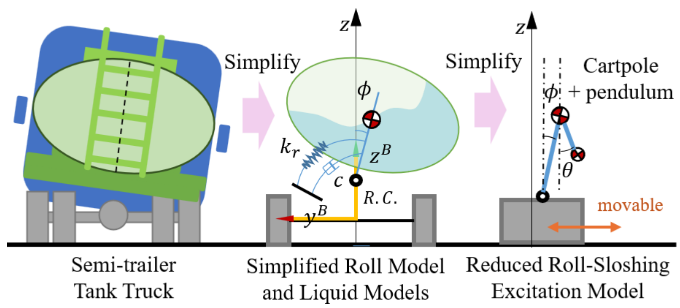

Fig. 2: Reduced lateral model. The online controller only needs the frequency response between input and output.

B. Critical Band and Risk Bound

Let the rollover-risk output be

$$z = C_z\mathbf{x}_r + D_z a_y, \tag{3}$$

where z represents LTR, roll angle, roll rate, or another rollover index, and in this work roll rate or a residual related to roll/yaw amplification is used as the measurable risk-output when available. From (1), the input-output frequency response is

$$G_{za}(s) = C_z(s^2\mathbf{M} + s\mathbf{C} + \mathbf{K})^{-1}\mathbf{B}_a + D_z. \tag{4}$$

The Q-band $\mathcal{B}_i = [f_i^-, f_i^+]$ is selected around the broad-peak region of $|G_{za}(j2\pi f)|$, where f_i^- and f_i^+ are the lower and upper frequencies of the ith band. In this work the main band is centered on the dominant sloshing-related response, with additional bands covering secondary and higher-frequency coupled excitations.

Safety could be stated as a sufficient frequency-domain condition. Let $a_{y,\mathcal{B}}$ denote the lateral-acceleration component in a target band $\mathcal{B}$, and let $z_{\mathcal{B}}$ be the corresponding risk-output component. Around the calibrated operating point, assume that the linearized system is stable and

$$|G_{za}(j2\pi f)| \le \bar{G}_{\mathcal{B}}, \qquad f \in \mathcal{B}. \tag{5}$$

Then Parseval's relation gives

$$\|z_{\mathcal{B}}\|_2^2 \le \bar{G}_{\mathcal{B}}^2 \|a_{y,\mathcal{B}}\|_2^2. \tag{6}$$

Thus, if $\|a_{y,\mathcal{B}}\|_2^2 \le \bar{E}_{\mathcal{B}}$, where $\bar{E}_{\mathcal{B}}$ is the given band-energy threshold,

$$\|z_{\mathcal{B}}\|_2 \le \bar{G}_{\mathcal{B}} \sqrt{\bar{E}_{\mathcal{B}}}. \tag{7}$$

Considering static load-transfer contribution and bounded residual terms, a conservative rollover margin can be written as

$$|\mathrm{LTR}_{\mathrm{stat}}| + \bar{G}_{\mathcal{B}} \sqrt{\bar{E}_{\mathcal{B}}} + R_\perp + R_0 + R_m \le \mathrm{LTR}_{\mathrm{lim}}, \tag{8}$$

where $R_\perp$, R_0, and R_m bound non-target-band excitation, initial roll/slosh energy, and model or sensing errors, respectively. Proof: Inequality (6) follows from the frequency-domain gain bound of the stable linearized input-output model. Substituting the band-energy threshold gives (7). Adding the bounded static, off-band, initial-condition, and model-error contributions yields (8). The condition is sufficient rather than necessary, and its conservatism comes from linearization, band discretization, gain upper bounding, and residual margins.

This proof explains why limiting the band energy of a_y can reduce rollover-risk excitation even when liquid phase and true wheel loads are not observed online. The discrete Q-band matrix used by the controller is an implementable approximation of $\|a_{y,\mathcal{B}}\|_2^2$ over the prediction horizon.

III. Adaptive Q-Band Suppression Controller

A. Tractor Prediction Model and ESO

The online optimizer uses a low-dimensional tractor-side nonlinear prediction model. Trailer force, liquid sloshing, tire mismatch, road slope, and parameter errors are not introduced as explicit states; instead, they are collected by residual terms. Let v_x, v_y, r, and δ denote tractor longitudinal speed, lateral speed, yaw rate, and front-wheel steering angle. With front and rear cornering stiffnesses C_f and C_r, the tire slip angles and lateral forces are computed as

$$\alpha_f = \delta - \frac{v_y + l_f r}{v_x}, \qquad \alpha_r = -\frac{v_y - l_r r}{v_x}, \tag{9}$$

$$F_{yf} = C_f \alpha_f, \qquad F_{yr} = C_r \alpha_r, \tag{10}$$

where l_f and l_r are the distances from tractor center of gravity to the front and rear axles. The nominal prediction model is

$$\begin{aligned} \dot{v}_y &= \frac{F_{yf}\cos\delta + F_{yr}}{m_{\mathrm{eq}}} - v_x r + d_y, \\ \dot{r} &= \frac{l_f F_{yf}\cos\delta - l_r F_{yr}}{I_{z,\mathrm{eq}}} + d_r, \end{aligned} \tag{11}$$

where m_{eq} and $I_{z,\mathrm{eq}}$ are the hitch-angle-scheduled equivalent lateral mass and yaw inertia, and d_y, d_r are lumped lateral and yaw residuals. The lateral acceleration used by Q-band is computed from the prediction model as

$$a_{y,k}^{\mathrm{pred}} = \dot{v}_{y,k} + v_{x,k} r_k. \tag{12}$$

An extended-state observer (ESO) estimates the yaw-channel residual:

$$\begin{aligned} \dot{\hat{r}} &= \frac{l_f \hat{F}_{yf}\cos\delta - l_r \hat{F}_{yr}}{I_{z,\mathrm{eq}}} + \hat{d}_r + \beta_{r1}(r - \hat{r}), \\ \dot{\hat{d}}_r &= \beta_{r2}(r - \hat{r}), \end{aligned} \tag{13}$$

where $\hat{r}$ and $\hat{d}_r$ are the estimated yaw rate and yaw residual, and $\beta_{r1}, \beta_{r2} > 0$ are observer gains. The lateral residual is obtained from the difference between the measured lateral acceleration and the kinematic centripetal component:

$$\hat{d}_y = a_y^{\mathrm{centri}} - a_y^{\mathrm{meas}} = v_x r - (a_y^{\mathrm{imu}} - \Delta a_{y,0}), \tag{14}$$

where a_y^{imu} is the IMU lateral acceleration, $\Delta a_{y,0}$ is its slowly updated zero offset, and $a_y^{\mathrm{centri}} = v_x r$ is the low-dimensional centripetal estimate. In the prediction horizon, the residual is held or decayed as

$$\hat{d}_{\ell|k} = \lambda_d^{\ell} \hat{d}_k, \qquad 0 < \lambda_d \le 1, \tag{15}$$

where ℓ is the prediction step and λ_d is the residual forgetting factor.

B. Hitch-Angle and Equivalent Inertia

The equivalent parameters in (11) require a tractor-trailer articulation prior. Let the tractor and trailer headings be ψ_1 and ψ_2, and define the hitch angle as

$$\gamma = \psi_1 - \psi_2. \tag{16}$$

With measured tractor yaw rate r_m, the trailer heading and hitch angle are estimated by

$$\begin{aligned} \dot{\hat{\psi}}_2 &= K_v \frac{v_x}{L_2}\left(\frac{L_h}{L_1}\tan\delta\cos\hat{\gamma} + \sin\hat{\gamma}\right), \\ \dot{\hat{\gamma}} &= r_m - \dot{\hat{\psi}}_2. \end{aligned} \tag{17}$$

Here L_1 is the tractor wheelbase, L_2 is the distance from the hitch to the trailer axle group, and L_h is the distance from the tractor rear axle to the hitch. The correction factor

$$K_v = 1 - \frac{m_2 v_x^2 e}{L_1 L_2 k_3} \tag{18}$$

approximates trailer-axle sideslip, where m_2 is trailer mass, e is the equivalent hitch-to-trailer geometric offset, and k_3 is the trailer axle cornering stiffness. When these trailer tire parameters are not available, $K_v = 1$ is used.

The trailer contribution is projected into tractor lateral inertia and yaw inertia:

$$m_{x,\mathrm{eq}}(\hat{\gamma}) = m_1 + m_2\cos^2\hat{\gamma}, \qquad m_{y,\mathrm{eq}}(\hat{\gamma}) = m_1 + m_2\sin^2\hat{\gamma}, \tag{19}$$

$$I_{z,\mathrm{eq}}(\hat{\gamma}) = I_{z1} + I_{z2}\left(\frac{d_1}{d_2}\right)^2 \cos^2\hat{\gamma} + m_2 d_1^2 \sin^2\hat{\gamma}. \tag{20}$$

Here m_1 is tractor mass, I_{z1} and I_{z2} are tractor and trailer yaw inertias, and d_1, d_2 are the equivalent geometric distances used to project trailer yaw inertia to the tractor yaw axis. In the NMPC model, $m_{\mathrm{eq}} = m_{y,\mathrm{eq}}(\hat{\gamma})$, and $I_{z,\mathrm{eq}}$ is evaluated from (20), after low-pass filtering, upper/lower saturation, and rate limiting.

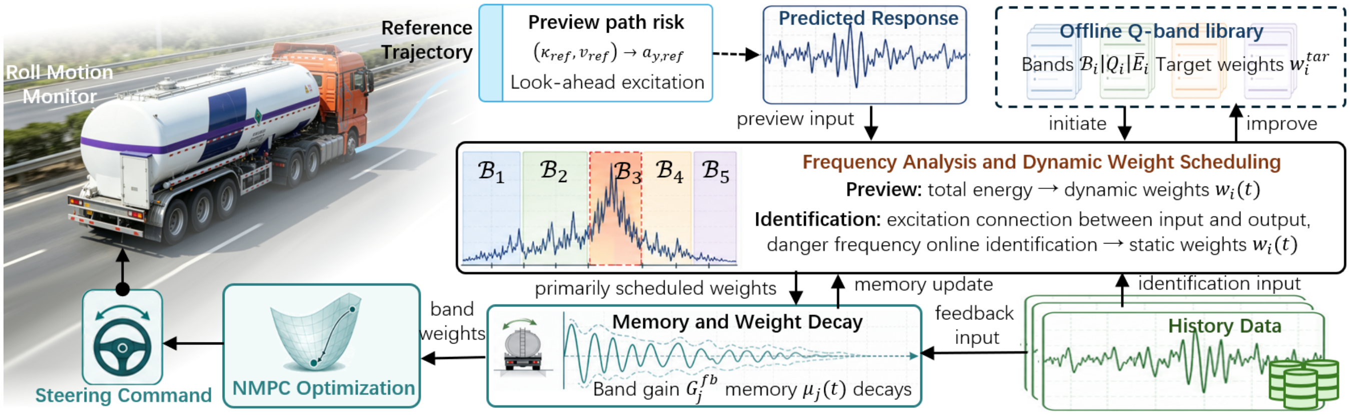

Fig. 3: Adaptive Q-band scheduling. Preview risk raises the default band weights before an aggressive path segment, while feedback band monitoring further increases the weights of dynamically amplified bands and delays weight release after residual sloshing is detected.

C. Band-Energy Projection

Let the prediction horizon contain N samples with sampling time Δt. The predicted lateral-acceleration sequence is

$$\mathbf{A}_y = [a_{y,0}, a_{y,1}, \dots, a_{y,N-1}]^{\mathrm{T}}, \tag{21}$$

where $a_{y,k}$ is computed from (12) for a dynamic predictor, or from $a_{y,k} \approx v_{x,k}^2 \kappa_k$ for a kinematic predictor with path curvature κ_k. For a critical band $\mathcal{B}_i = [f_i^-, f_i^+]$, choose discrete frequencies f_{ij} and time grid $t_k = k\Delta t$. Define

$$\mathbf{c}_{ij} = [\cos(2\pi f_{ij} t_0), \dots, \cos(2\pi f_{ij} t_{N-1})]^{\mathrm{T}}, \tag{22}$$

$$\mathbf{s}_{ij} = [\sin(2\pi f_{ij} t_0), \dots, \sin(2\pi f_{ij} t_{N-1})]^{\mathrm{T}}. \tag{23}$$

The predicted energy in the ith band is

$$E_i(\mathbf{A}_y) = \mathbf{A}_y^{\mathrm{T}} \mathbf{Q}_i \mathbf{A}_y, \tag{24}$$

$$\mathbf{Q}_i = \frac{1}{N^2} \sum_j (\mathbf{c}_{ij}\mathbf{c}_{ij}^{\mathrm{T}} + \mathbf{s}_{ij}\mathbf{s}_{ij}^{\mathrm{T}}) \succeq 0. \tag{25}$$

The matrix $\mathbf{Q}_i$ is precomputed from Δt, N, and $\mathcal{B}_i$. It adds no vehicle state; it only projects the predicted input sequence $\mathbf{A}_y$ onto dangerous frequency components.

D. Optimization Problem

Let $\mathbf{x}_k$ be the prediction state, $u_k = \delta_{\mathrm{cmd},k}$ be the steering command, $\mathbf{U} = [u_0, \dots, u_{N-1}]^{\mathrm{T}}$, and $\mathbf{S} = [s_1, \dots, s_{n_b}]^{\mathrm{T}}$ be the nonnegative slack variables for n_b Q-bands. The Q-band ART objective is

$$\begin{aligned} \min_{\mathbf{U},\mathbf{S}} \; & J_{\mathrm{trk}} + J_u + J_{\Delta u} + w_a \|\mathbf{A}_y\|_2^2 + w_1 \|\mathbf{D}_1 \mathbf{A}_y\|_2^2 \\ & + w_2 \|\mathbf{D}_2 \mathbf{A}_y\|_2^2 + \sum_{i=1}^{n_b} w_i(t) \mathbf{A}_y^{\mathrm{T}} \mathbf{Q}_i \mathbf{A}_y + \sum_{i=1}^{n_b} \rho_i s_i^2. \end{aligned} \tag{26}$$

Here $J_u = w_u \|\mathbf{U}\|_2^2$, $J_{\Delta u} = w_{\Delta u} \|\mathbf{D}_u \mathbf{U}\|_2^2$, and w_a, w_1, w_2, ρ_i are positive tuning weights. The first- and second-order difference matrices are calculated by

$$(\mathbf{D}_1 \mathbf{A}_y)_k = \frac{a_{y,k+1} - a_{y,k}}{\Delta t}, \tag{27}$$

$$(\mathbf{D}_2 \mathbf{A}_y)_k = \frac{a_{y,k+2} - 2a_{y,k+1} + a_{y,k}}{\Delta t^2}. \tag{28}$$

The tracking cost uses the previewed reference position and heading:

$$J_{\mathrm{trk}} = w_{xy}(\|\mathbf{X} - \mathbf{X}_{\mathrm{ref}}\|_2^2 + \|\mathbf{Y} - \mathbf{Y}_{\mathrm{ref}}\|_2^2) + w_\psi \|\boldsymbol{\psi} - \boldsymbol{\psi}_{\mathrm{ref}}\|_2^2, \tag{29}$$

where $\mathbf{X}$, $\mathbf{Y}$, and $\boldsymbol{\psi}$ collect predicted tractor positions and headings over the horizon, and the subscript "ref" denotes the path-preview sequence.

The constraints are

$$\begin{aligned} \mathbf{x}_{k+1} &= f(\mathbf{x}_k, u_k, \hat{d}_{k|k}, m_{\mathrm{eq}}, I_{z,\mathrm{eq}}), \\ |a_{y,k}| &\leq a_{y,\max}^{\mathrm{dyn}}, \\ u_{\min} \leq u_k &\leq u_{\max}, \quad |\Delta u_k| \leq \Delta u_{\max}, \\ E_i(\mathbf{A}_y) &\leq \bar{E}_i + s_i, \quad s_i \geq 0. \end{aligned} \tag{30}$$

The bound $a_{y,\max}^{\mathrm{dyn}}$ is the speed- and risk-dependent lateral-acceleration limit, and $\bar{E}_i$ is the calibrated safe energy threshold of band $\mathcal{B}_i$.

E. Preview-Adaptive Scheduling

Q-band ART uses preview risk as the main trigger. The reference lateral-acceleration sequence $\mathbf{A}_{y,\mathrm{ref}}$ is computed from future path curvature as

$$a_{y,\mathrm{ref},k} = v_{x,k}^2 \kappa_{\mathrm{ref},k}, \qquad \kappa_{\mathrm{ref},k} \approx \frac{\mathrm{wrap}(\psi_{\mathrm{ref},k+1} - \psi_{\mathrm{ref},k})}{\Delta s_k}, \tag{31}$$

where Δs_k is the preview arc-length interval and $\mathrm{wrap}(\cdot)$ keeps heading differences within $[-\pi, \pi]$. The preview risk in band i is

$$R_i^{\mathrm{pre}} = \left[\frac{\mathbf{A}_{y,\mathrm{ref}}^{\mathrm{T}} \mathbf{Q}_i \mathbf{A}_{y,\mathrm{ref}}}{\bar{E}_{i,\mathrm{ref}}} - 1 \right]_+, \tag{32}$$

where $[x]_+ = \max(x, 0)$ and $\bar{E}_{i,\mathrm{ref}}$ is the preview activation threshold. The preview-only activation is

$$\lambda_i^{\mathrm{pre}} = 1 - \exp(-\kappa_{\mathrm{pre}} R_i^{\mathrm{pre}}), \tag{33}$$

where $\kappa_{\mathrm{pre}} > 0$ controls how quickly the weight rises before an aggressive maneuver.

F. Feedback Band Monitoring and Residual Sloshing Memory

Preview alone cannot identify a frequency that becomes dangerous because of current loading, liquid fill level, tire state, or unmodeled trailer dynamics. Therefore, a feedback monitor is added to identify bands in which the input exists and the measured output is significantly amplified. Over a sliding window of M samples, define the input vector

$$\mathbf{a}_M(t) = [a_y^{\mathrm{mon}}(t-M+1), \ldots, a_y^{\mathrm{mon}}(t)]^{\mathrm{T}}, \qquad (34)$$

where a_y^{mon} is the measured or predicted lateral acceleration after offset correction. The monitored output is

$$y^{\mathrm{mon}} \in \{r - \hat{r},\ a_y^{\mathrm{imu}} - a_y^{\mathrm{pred}},\ \dot{\phi}\}, \qquad (35)$$

with the output window

$$\mathbf{y}_M(t) = [y^{\mathrm{mon}}(t-M+1), \ldots, y^{\mathrm{mon}}(t)]^{\mathrm{T}}. \qquad (36)$$

The yaw residual and acceleration residual are available from tractor-side sensing, and roll rate $\dot{\phi}$ is used when a roll sensor exists on the validation platform. For a candidate monitoring band $\mathcal{C}_j$, construct $\mathbf{Q}_j^M$ in the same way as $\mathbf{Q}_i$ but using window length M. The input and output band energies are

$$E_{a,j}^{\mathrm{fb}} = \mathbf{a}_M^{\mathrm{T}} \mathbf{Q}_j^M \mathbf{a}_M, \qquad E_{y,j}^{\mathrm{fb}} = \mathbf{y}_M^{\mathrm{T}} \mathbf{Q}_j^M \mathbf{y}_M, \qquad (37)$$

and the observed amplification is

$$G_j^{\mathrm{fb}} = \sqrt{\frac{E_{y,j}^{\mathrm{fb}}}{E_{a,j}^{\mathrm{fb}} + \epsilon}}, \qquad (38)$$

where ϵ avoids division by zero. A dynamically identified dangerous band must satisfy both non-negligible input and amplified output, which is encoded by

$$R_j^{\mathrm{fb}} = \left[\frac{E_{a,j}^{\mathrm{fb}}}{\bar{E}_{a,j}} - 1\right]_+ \left[\frac{G_j^{\mathrm{fb}}}{\bar{G}_j^{\mathrm{fb}}} - 1\right]_+. \qquad (39)$$

Here $\bar{E}_{a,j}$ is the minimum input-energy threshold for declaring that the band is excited, and $\bar{G}_j^{\mathrm{fb}}$ is the nominal amplification level obtained from calibration or low-risk data.

To remember residual sloshing after the steering input has decreased, the feedback risk is stored as

$$\mu_j(t) = \max\{\lambda_m \mu_j(t-\Delta t),\ R_j^{\mathrm{fb}}(t)\}, \qquad 0 < \lambda_m < 1, \qquad (40)$$

where λ_m is the memory decay factor. If the monitoring bands $\mathcal{C}_j$ are finer than the default controller bands $\mathcal{B}_i$, the feedback memory is mapped to the controller band by the overlap ratio

$$\mu_i^{\mathrm{dyn}}(t) = \max_j \sigma_{ij} \mu_j(t), \qquad \sigma_{ij} = \frac{|\mathcal{B}_i \cap \mathcal{C}_j|}{|\mathcal{B}_i|}. \qquad (41)$$

Thus, bands that are observed to be both excited and amplified act as dynamically identified dangerous bands and increase the default Q-band weights.

The preview and feedback activations are fused before filtering:

$$\lambda_i(t) = 1 - \exp[-\kappa_{\mathrm{pre}} R_i^{\mathrm{pre}} - \kappa_{\mathrm{fb}} \mu_i^{\mathrm{dyn}}(t)], \qquad (42)$$

$$w_{i,\mathrm{tar}}^{\mathrm{dyn}}(t) = \min\{w_i^{\max},\ w_i^{\mathrm{tar}}(1 + \eta_{\mathrm{fb}} \mu_i^{\mathrm{dyn}}(t))\}, \qquad (43)$$

$$w_i^{\mathrm{raw}}(t) = w_i^{\min} + \lambda_i(t) \left(w_{i,\mathrm{tar}}^{\mathrm{dyn}}(t) - w_i^{\min}\right). \qquad (44)$$

Here κ_{fb} and η_{fb} tune the feedback contribution, $w_i^{\min}$ is the low-risk weight, w_i^{tar} is the nominal calibrated target weight, and $w_i^{\max}$ prevents excessive conservatism. Finally, a non-symmetric filter is applied:

$$w_i(t) = (1-\alpha_i) w_i(t-\Delta t) + \alpha_i w_i^{\mathrm{raw}}(t), \qquad (45)$$

where $\alpha_i = \alpha_\uparrow$ when $w_i^{\mathrm{raw}} > w_i(t-\Delta t)$ and $\alpha_i = \alpha_\downarrow$ otherwise, with $\alpha_\uparrow > \alpha_\downarrow$. This makes the weight rise quickly when preview or feedback detects risk, and decay slowly while residual sloshing memory remains.

G. Implementation Notes

The Q-band module is model-agnostic. For a kinematic predictor, $a_{y,k}$ can be approximated from curvature or steering; for a dynamic predictor, it can be computed from $\dot{v}_y + v_x r$; for TruckSim or model-truck validation, it is evaluated from the predicted tracking response. In all cases, the controller suppresses the lateral excitation sequence rather than explicitly controlling the liquid state.

IV. Validation

A. Co-Simulation Setup

The simulation validation was conducted on the TruckSim–STAR-CCM+ vehicle–liquid coupling platform reported in [27]. The vehicle is the default full-load 49-ton 3A sleeper tractor with trailer. The tank is filled to 50%, and the liquid–vehicle interaction is generated by a three-dimensional STAR-CCM+ sloshing model coupled with the nonlinear TruckSim vehicle model. This high-fidelity platform is used only for validation; the proposed Q-band controller does not use the liquid states, trailer-side states, or real-time fluid forces as feedback.

The main comparison uses one continuous five-pins maneuver. The longitudinal speed increases linearly from 60 km/h to 100 km/h within 60 s, covering the lower and upper highway speed limits for trucks. The five double-lane-change (DLC) maneuvers become progressively more aggressive: the two lane changes within DLC1–DLC5 are completed over longitudinal distances of 100, 81.25, 62.5, 43.75, and 25 m, respectively. Therefore, the last pins simultaneously impose higher speed, shorter lane-change distance, and stronger lateral excitation.

Three controllers are compared. The first is the built-in TruckSim preview tracking (PT), which tracks the reference path without considering rollover risk. The second is the full-state active rollover tracking controller (Full-ART) from [27], which explicitly uses the trailer and liquid

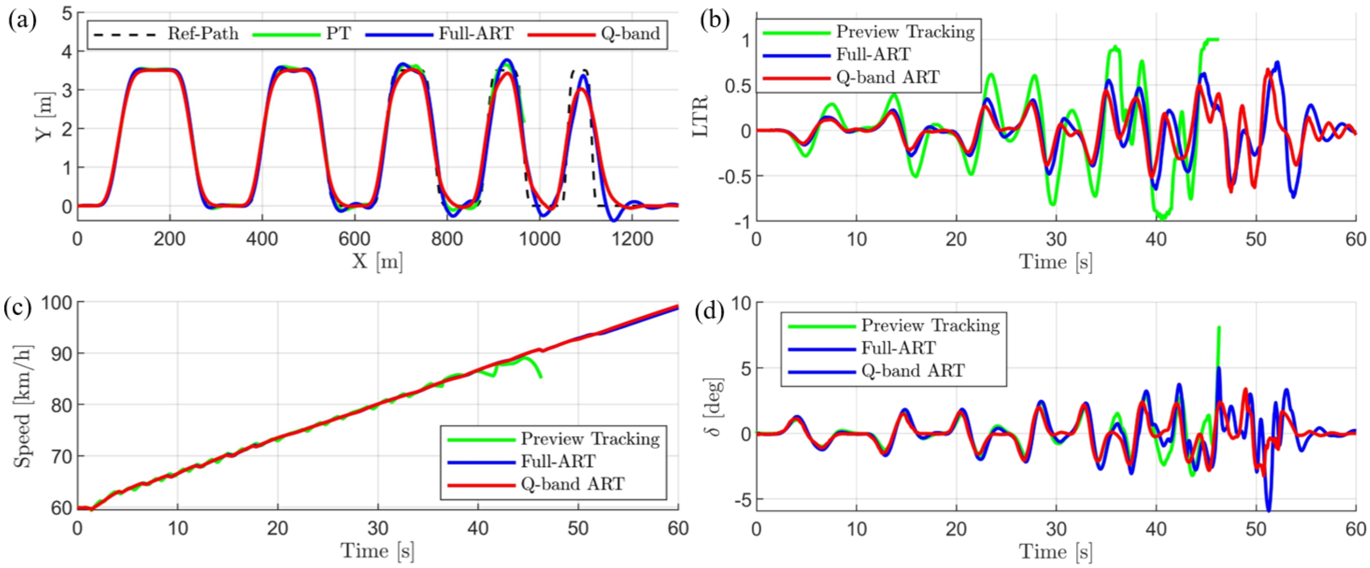

Fig. 4: Continuous five-pins comparison: (a) trajectory tracking, (b) lateral load transfer ratio, (c) longitudinal speed, and (d) front steering angle. PT denotes the TruckSim preview tracking baseline, Full-ART denotes the full-state active rollover tracking controller in [27], and Q-band ART denotes the proposed controller.

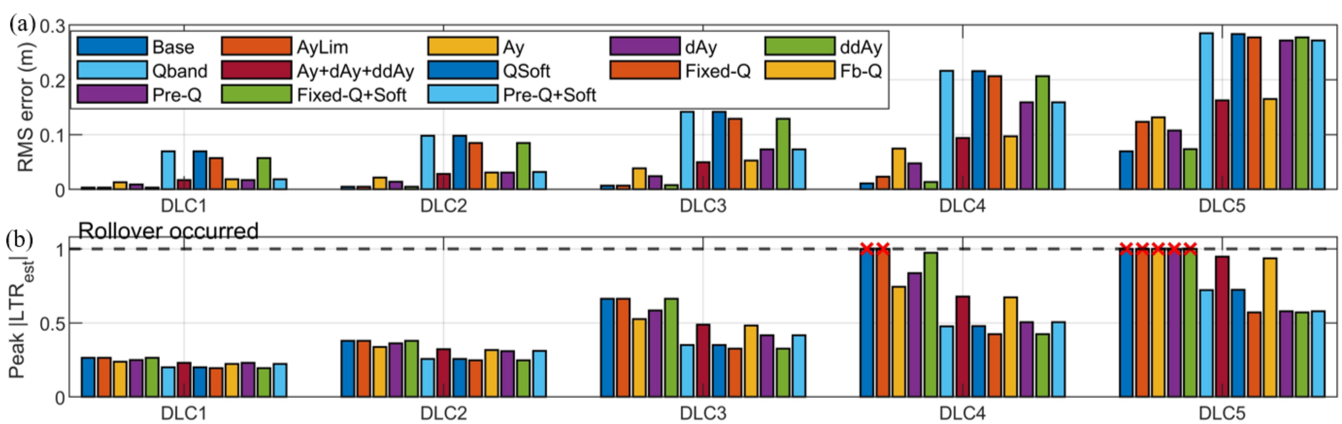

Fig. 5: Ablation study of the Q-band mechanism in five independent DLC segments: (a) RMS tracking error and (b) peak estimated LTR. Red crosses at the dashed $|\mathrm{LTR}_{\mathrm{est}}| = 1$ line indicate simulations in which rollover occurred, not numerical clipping.

dynamics and requires liquid-state observation together with trailer-state feedback. This method can drive the vehicle closer to its physical handling limit, but its online model, calibration, and sensing requirements are high. The third is the proposed Q-band ART, which uses only tractor-side motion information and suppresses dangerous lateral-excitation frequency bands. The comparison is intended to show whether a low-observability controller can approach the anti-rollover performance of the full-state method.

B. Continuous Five-Pins Results

Figure 4 shows the continuous five-pins comparison. PT follows the first several pins tightly, but it keeps pursuing the geometric path after the maneuver becomes dynamically unsafe. Its LTR increases rapidly in the later pins and reaches the rollover boundary at approximately 46 s, after which the simulation cannot continue as a physically valid run. This confirms that pure preview tracking is inadequate for a high-speed liquid tank semi-trailer, even when the reference path itself is smooth.

Full-ART and Q-band ART both complete the 60 s maneuver without rollover. Full-ART achieves slightly tighter path tracking in the most aggressive pins because it uses a richer vehicle–liquid prediction model and full-state feedback. In contrast, Q-band ART intentionally relaxes the path in high-risk portions, especially near the fifth pin, so that the lateral excitation does not continue feeding the dominant roll–sloshing bands. The resulting trajectory is slightly more conservative, but the LTR response remains comparable to that of Full-ART for most of the maneuver. The speed curves of Full-ART and Q-band ART nearly overlap, indicating that the comparison is not biased by different speed histories. The steering-angle response also remains within practical bounds; Q-band ART reduces the most dangerous oscillatory steering content rather than relying on abrupt large steering reversals.

These results highlight the main tradeoff of the proposed method. Full-ART can exploit a detailed trailer–liquid model to approach the vehicle limit more aggressively, but this requires information that is difficult to obtain on production vehicles. Q-band ART sacrifices a small amount of geometric tracking tightness and instead obtains a similar rollover-prevention effect using only simple tractor-side measurements and a low-order prediction model.

C. Q-Band Ablation Study

The ablation study is conducted with five independent DLC simulations rather than one continuous trajectory. This choice is necessary because, once a controller rolls the vehicle over in an earlier pin, the subsequent pins are no longer physically meaningful. For each DLC, the initial speed and pose are reset to the corresponding rollover-free Q-band condition at the entrance of the segment. The compared variants include tracking only (Base), lateral-acceleration amplitude limitation (AyLim), acceleration magnitude and difference penalties, fixed Q-band weighting, feedback-adjusted Q-band weighting, preview Q-band weighting, and soft-constrained Q-band combinations.

Figure 5 summarizes the ablation results. In DLC1 and DLC2, all variants are far from the rollover boundary, and the tracking errors remain small. This indicates that the additional Q-band terms do not unnecessarily dominate the controller in low-risk maneuvers. As the lane-change distance decreases in DLC3, the difference between amplitude-based regulation and frequency-selective regulation becomes visible. The variants that include Q-band weighting reduce the peak $|\mathrm{LTR}_{\mathrm{est}}|$ while keeping the RMS tracking error within an acceptable range.

The difference becomes decisive in DLC4 and DLC5. The Base controller and several acceleration-only variants reach the rollover boundary. Importantly, AyLim uses the same lateral-acceleration amplitude limitation, yet it still rolls over in the severe segments. Therefore, rollover prevention cannot be explained by limiting only the peak value of a_y. Even when the acceleration magnitude is bounded, repeated excitation near the dominant roll–sloshing bands can accumulate energy and trigger rollover. This observation directly supports the Q-band theory: the dangerous part is not only the acceleration amplitude, but also its frequency content over the preview horizon.

Among the Q-band variants, fixed Q-band weighting already reduces the peak LTR in the severe DLCs, showing that the frequency-selective penalty is the core contributor. The soft-constrained variants further improve robustness by allowing controlled tracking relaxation when the maneuver approaches the roll limit. Preview-adaptive weighting is particularly useful in DLC4 and DLC5 because it increases the dangerous-band penalty before the vehicle enters the strongest lane-change excitation. Feedback adjustment and residual memory provide an additional safety margin when the measured response shows amplified output in a band that is being excited by the steering input. Overall, the ablation confirms that the main protective mechanism is the suppression of lateral excitation energy in the dangerous frequency bands, while preview scheduling and soft constraints determine how early and how smoothly this protection is activated.

TABLE I: Validation Summary With Replaceable Results

Case	Rollover result	Tracking result	Online information
PT simulation	rollover at 46 s	tight before rollover	tractor states
Full-ART simulation	no rollover	tightest near limit	tractor, trailer, and liquid states
Q-band ART simulation	no rollover	slightly conservative	tractor states only
Model-truck baseline	xxx	xxx	–
Model-truck Q-band ART	xxx	xxx	xxx ms

D. Model-Truck Experiment

The 1:14 semi-trailer tank truck experiment was conducted to verify that the Q-band mechanism remains effective on a physical platform. The experiment used a double-lane-change or 5-pins-like reference path scaled according to the model-truck operating envelope. Baseline tracking and Q-band ART were compared under the same speed and filling-rate settings. The measured responses include path tracking, steering command, lateral acceleration, roll/yaw response, liquid-surface motion when available, and controller solve time.

The model-truck results show the same qualitative behavior as the simulation. Baseline tracking follows the reference more tightly but excites larger roll-slosh motion, and the anti-roll support/contact or rollover-risk proxy reaches xxx. Q-band ART reduces the peak risk proxy by xxx%, keeps the liquid/roll response within xxx, and increases RMS tracking error from xxx m to xxx m. The controller average and peak solve times are xxx ms and xxx ms, respectively, satisfying the real-time requirement of the experimental platform.

V. Conclusion

This paper proposed Q-band ART, an adaptive frequency-band suppression method for anti-rollover tracking of semi-trailer tank trucks. Instead of requiring liquid sloshing states or wheel-load-based LTR as online controller states, the method suppresses the predicted lateral-acceleration energy in critical vehicle-liquid bands. A reduced roll-sloshing model provides the frequency-domain safety interpretation, and the Q-band term is integrated with acceleration smoothing, soft constraints, preview-adaptive scheduling, ESO disturbance compensation, and hitch-angle-based parameter scheduling.

The 5-pins simulation and 1:14 model-truck experiment show that Q-band ART reduces peak rollover-risk proxy from xxx to xxx and improves RMS LTR/risk response by xxx%, while keeping tracking error and computation time within acceptable ranges. The results indicate that input-side critical-band suppression is a practical complement to explicit rollover-state constraints, especially when trailer-side sensing and liquid-state observation are limited. Future work will refine critical-band calibration across filling rates and integrate Q-band scheduling with combined steering and braking control.

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